



AEGIS

Anonymization & Exchange Gateway
for Imaging Studies

All DICOM Modalities

MRI CT PET US XR NM MG RT

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Anonymization & Exchange Gateway for Imaging Studies

A multi-cloud platform for secure, HIPAA-compliant de-identification and sharing of medical imaging data — for research teams and radiology departments alike

In Greek mythology, the aegis was the divine shield of Zeus and Athena — a symbol of protection. The name captures our mission: shielding patient identity while enabling the free flow of imaging data for research and clinical care.

The Problem

Hospitals and research institutions need to share medical imaging data — MRI, CT, PET, ultrasound, X-ray — for multi-site clinical trials, research collaborations, and second opinions. Doing this safely is harder than it should be.

De-identification tools frequently fail. A 2015 study in *European Radiology* tested 10 free DICOM de-identification tools against 50 required PHI data elements. With default settings, only 1 of 10 tools successfully removed all required elements; 4 tools achieved success rates of 26% or less.^[1]

Metadata removal alone is not enough for brain imaging. A 2019 *New England Journal of Medicine* correspondence from Mayo Clinic showed that automated face-recognition software correctly matched de-identified brain MRI participants to their photographs in **83% of cases** — using only the scan geometry, no metadata at all.^[2] This means head imaging requires an additional step: automated defacing (removal of facial surface features from the 3D volume).

Burned-in PHI is invisible to tag-level tools. Patient names, dates of birth, and accession numbers are routinely overlaid directly on image pixels — particularly in ultrasound, CT topograms, and fluoroscopy. Standard DICOM de-identification tools operate on metadata only and miss this entirely.^[3]

Installing software at sending sites is a barrier. Platforms like XNAT require a Java-based desktop client or server daemon installed and IT-approved at every participating hospital. MIRC CTP requires a standalone Java application. Each new site adds months of IT negotiation.

Data sharing is now federally mandated — but tooling lags. The NIH Data Management and Sharing Policy (NOT-OD-21-013, effective January 25, 2023) requires all NIH-funded investigators to share scientific data as a condition of their grant award, no later than the time of first publication.^[4] Institutions are now obligated to share data they have no reliable, low-friction way to de-identify and transmit.

Result: Multi-site studies are delayed by months. In a 2021 study of global randomized trials, contract execution alone averaged 7.9 months per US site.^[5] Software installation and IT approval add to this.

Market Opportunity

The medical image exchange market is growing, driven by federal data sharing mandates and expanding multi-site clinical trial volume in oncology and neurology.

Market	2024 Estimate	2034 Projection	CAGR
Medical Image Exchange Systems ^[6]	\$3.9B	\$9.9B	9.7%

Why now:

- The NIH DMS Policy (2023) creates a compliance obligation for tens of thousands of active grants that did not previously require a data sharing plan
- A 2024 NCI-sponsored benchmark (MIDI-B, presented at MICCAI 2024) formally tested de-identification tools and found that burned-in pixel PHI and free-text fields remain the hardest unsolved problems — no existing tool handles all cases reliably^[7].
- Growing multi-site trial volume in oncology (PSMA PET, amyloid PET, whole-body MRI) and neurology (tau PET, fMRI) is driving demand for scalable data collection from distributed hospital networks
- Cloud-hosted DICOM storage (GCP Healthcare API, AWS HealthImaging) is now mature commodity infrastructure, reducing the barrier to building managed platforms

The Solution

AEGIS is a cloud-hosted platform that makes secure medical image sharing as easy as uploading a file — for any DICOM modality. It serves two complementary audiences from a single platform.

Capability	What It Does
No software to install	Upload portal runs entirely in the web browser. Hospitals use their existing computers and internet connection — no desktop app, no IT approval for new software.
Tag-level anonymization in the browser	Patient data is stripped from DICOM tags before it leaves the hospital network, following the DICOM PS3.15 Annex E Basic Confidentiality Profile — the established international standard covering all 18 HIPAA Safe Harbor identifier categories. ^[8]
Automated defacing for head scans	MRI, CT, and PET studies of the head and brain undergo server-side defacing (removal of facial surface geometry) before entering the research archive. Non-head studies bypass this step automatically.

Burned-in PHI detection	Automated OCR scans image pixels for overlaid text (patient names, dates, accession numbers) that tag-level tools miss. Studies with detected pixel PHI are flagged for review.
Supports all DICOM modalities	MRI, CT, PET, PET/CT, ultrasound, X-ray, mammography, nuclear medicine, and more — using the same upload and anonymization workflow.
MRI protocol compliance	Automated verification that acquisition parameters (TR, TE, flip angle, resolution) match site-specific templates per scanner manufacturer, model, and software version — catching the 0.19–64% non-compliance rates found across multi-site studies. [10]
Centralized audit trail	Every upload, approval, rejection, and data export is logged with a timestamp and actor. Institutions can demonstrate HIPAA compliance from a single dashboard.
Clinician review before release	Administrators review anonymized images in a web-based DICOM viewer (OHIF) before approving studies for sharing. Nothing is shared automatically without human sign-off.

Who It's For

AEGIS serves two audiences from a single de-identification and routing platform:

Multi-site research teams need to collect imaging data from hospitals they don't control, de-identify it to meet NIH and IRB requirements, and convert it to research-ready formats. AEGIS gives them a browser link they can send to any participating site — no software installation, no IT negotiation. The research pipeline (protocol compliance, automated QC, BIDS conversion, defacing) ensures data arrives analysis-ready.

Hospital radiology departments need to de-identify imaging data for internal uses — training AI models, building real-world evidence databases, sharing with referring physicians, and fulfilling data requests from payers and registries. AEGIS gives them a centralized de-identification gateway with a complete audit trail, burned-in PHI detection, and configurable routing rules — without replacing their PACS or requiring a separate enterprise integration project.

	Research Teams	Radiology Departments
Entry point	Browser upload link sent to external sites	DIMSE receive / internal batch import

Key need	NIH DMS compliance, multi-site consistency	HIPAA de-identification, audit trail
Differentiator	Zero-install at sending sites	No PACS/VNA integration required
Pipeline features	Defacing, protocol compliance, QC, BIDS	Burned-in PHI, routing rules, export shares
Review workflow	PI/coordinator approves before archive	Radiologist/admin approves before release

The underlying platform — de-identification engine, routing rules, audit trail, OHIF viewer, and admin dashboard — is shared. Research-specific features (defacing, BIDS, protocol templates) and enterprise features (DIMSE receive, HL7/FHIR notifications, tag standardization) extend it for each audience.

How De-identification Works

AEGIS follows a two-phase approach that addresses the limitations documented in the literature:

Phase 1 – In the browser (before upload):

- └─ DICOM tags stripped per PS3.15 Annex E Basic Profile [\[8\]](#)
- └─ 18 HIPAA Safe Harbor identifiers addressed [\[9\]](#)
- └─ Reviewer sees before/after tag comparison before confirming
- └─ Only de-identified data is transmitted – PHI never leaves the hospital

Phase 2 – On the server (after upload):

- └─ De-identification completeness check
- └─ OCR scan of image pixels for burned-in text [\[3\]](#)
- └─ Head imaging → automated defacing [\[2\]](#)
- └─ Administrator review and approval before data is shared

This two-phase design directly addresses the gaps identified in the Aryanto (2015) [\[1\]](#) and Schwarz (2019) [\[2\]](#) studies.

Supported Modalities

Category	Modalities
Cross-sectional	MRI, CT, PET, PET/CT, PET/MR
Projection	X-Ray (CR/DR), Mammography (FFDM/DBT), Fluoroscopy
Nuclear Medicine	SPECT, Planar Scintigraphy
Ultrasound	Echocardiography, Abdominal, Vascular, OB/GYN
Radiation Therapy	RT Structure, RT Plan, RT Dose, RT Image
Other DICOM	Secondary Capture, Structured Reports, Key Images

Initial focus: Brain MRI, PET, and CT — these exercise the full pipeline including defacing. All other modalities use the same upload and de-identification workflow.

Technology

Layer	Technology	Notes
Backend	Go on Cloud Run (GCP) or ECS Fargate (AWS)	Compiled binary, minimal dependencies, fast cold starts
Frontend	React + TypeScript	Runs in any modern browser, no installation required
Database	PostgreSQL 15 (Cloud SQL / RDS)	Audit trail, project/user management, routing rules
DICOM Storage	Cloud-neutral (GCS, S3, or local filesystem)	Abstracted behind a pluggable storage interface
OCR / PHI Detection	Tesseract OCR (local) / Document AI (cloud)	Detects burned-in text in image pixels
Defacing	mri_deface (FreeSurfer)	Established tool, used in neuroimaging research
DICOM Viewer	OHIF Viewer (v3)	Open-source, browser-based, supports all modalities
Auth	GCP IAP / AWS ALB+Cognito / Azure AD	Multi-provider auth middleware, auto-detection

Infrastructure	Terraform (GCP + AWS modules)	Reproducible, version-controlled, multi-cloud
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On the use of automated tools: AEGIS uses automated tools to assist with — not replace — human review. Automated de-identification flags potential issues; a trained administrator reviews and approves every study before it is shared. Automated defacing quality is reviewed side-by-side against the original in the admin interface.

Compliance Framework

AEGIS is designed around established regulatory and technical standards:

Standard	Reference	What It Covers
HIPAA Privacy Rule	45 CFR § 164.514(b)	De-identification methods: Safe Harbor (18 identifiers) and Expert Determination
DICOM Confidentiality Profile	NEMA PS3.15 Annex E	Tag-level de-identification actions for all DICOM attributes
NIH Data Management Policy	NOT-OD-21-013 (eff. Jan 25, 2023)	Data sharing obligations for NIH-funded investigators
GCP Security	BAA, VPC, CMEK	Business Associate Agreement, encrypted storage, network controls

AEGIS does not create a new de-identification standard — it implements the ones that already exist.

Competitive Landscape

	AEGIS	ENCOG (Enlitic)	XNAT	Flywheel	MIRC CTP
Hosting	Multi-cloud (GCP, AWS, Azure)	On-premises / hybrid	Self-hosted	Commercial SaaS	On-premises

Install at sending site	None — browser only	PACS/VNA integration	Desktop Java client	CLI tool	Java application
Burned-in PHI	OCR detection	AI CV detection	None	Built-in	None
Defacing	Automated, reviewable	None	Manual or plugin	Built-in	None
Modalities	All DICOM	MR, CT, XR, US	Neuroimaging focus	All imaging	All imaging
Open source	Yes	No (commercial)	Yes	No	Yes
Audit trail	Centralized, per-study	Chain of custody	Per-instance	Built-in	Limited

Existing platforms tend to serve either research (XNAT, Flywheel) or enterprise radiology (ENCOG/Enlitic, MIRC CTP) — but not both. AEGIS is designed for both audiences from a shared platform, and is the only option that runs on any major cloud provider without re-architecture.

Enlitic's ENCOG is the closest commercial analogue to AEGIS's de-identification pipeline.^[12] It uses AI-driven computer vision to detect burned-in text overlays and claims to protect over 4,000 DICOM fields. However, ENCOG requires PACS/VNA integration at each sending site, does not offer volumetric defacing for head imaging, and publicly documents only four modalities (MR, CT, XR, ultrasound). ENCOG is part of a broader commercial suite (Ensign) that includes data standardization (ENDEX, FDA 510(k) cleared) and migration tools — but lacks the research pipeline features (protocol compliance, QC, BIDS) that multi-site studies require.

XNAT and Flywheel serve research well but require software installation at sending sites and offer limited enterprise radiology integration. MIRC CTP is a robust open-source pipeline but requires on-premises Java infrastructure and has no burned-in PHI detection or defacing.

Key differences from existing platforms:

1. **No software installation at sending sites** — the full anonymization workflow runs in the browser. ENCOG, XNAT, and MIRC CTP all require software deployed at the sending institution.
2. **Two-phase de-identification** — both tag-level (browser) and pixel-level (server-side OCR) are addressed in a unified pipeline
3. **Defacing is automated and reviewable** — not optional, manual, or absent. Neither ENCOG nor MIRC CTP offer volumetric defacing.
4. **Not specialty-specific** — designed for all DICOM modalities from the start, unlike XNAT (neuroimaging focus) or ENCOG (4 listed modalities)

5. **Dual-market platform** — research features (protocol compliance, QC, BIDS) and enterprise features (DIMSE receive, routing rules, tag standardization) extend a shared de-identification core. Competitors serve one audience or the other.
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Phased Roadmap

*Development status note: Phases 1–4 are **code-complete** — all features are implemented, compile, and pass CI. The platform has not yet been deployed to a cloud environment or tested with real clinical data. A test deployment on GCP or AWS is planned as the next milestone.*

Phase 1 — Foundation (Code Complete)

- Cloud infrastructure (Terraform for GCP and AWS)
- Browser-based upload portal with DICOM tag anonymization and before/after preview
- Go API with DICOM ingest and PostgreSQL audit trail
- Admin dashboard with OHIF viewer for QC, approve/reject workflow

Phase 2 — Defacing + Notifications (Code Complete)

- Automated mri_deface pipeline for head imaging
- Side-by-side before/after defacing review in admin interface
- Email notifications for upload confirmation, approval, and rejection

Phase 3 — Operations (Code Complete)

- Multi-project routing engine with configurable rules by modality, source, and anatomy
- Institution management with project-scoped roles
- Configurable anonymization profiles per project (which tags to retain for research)
- Audit log viewer with filtering
- Admin user management, RBAC enforcement (admin/viewer roles), authentication middleware

Phase 4 — Advanced Processing (Code Complete)

- Burned-in PHI detection using OCR on image pixels (addresses gap identified in [\[3\]](#), [\[7\]](#))

- Automated image quality assessment (motion artifact detection, coverage completeness)
- MRI protocol compliance — per-scanner, per-sequence parameter validation against configurable templates with tolerances, following the approach used by ADNI^[11] and tools like mrQA^[10]
- BIDS format conversion for neuroimaging research output
- Batch import tools for historical data migration
- Multi-cloud support — AWS S3 storage backend, ALB + Cognito auth, Terraform AWS module

Next Milestone — Test Deployment

- Deploy to GCP or AWS using Terraform modules
- End-to-end testing with sample DICOM datasets (TCIA public data)
- Validate full pipeline: upload → de-identify → deface → QC → approve → export
- Performance benchmarking and cost validation

Phase 5 — Enterprise Radiology

- DIMSE receive endpoint — hospitals push studies directly from PACS/VNA without browser upload
- DICOM tag standardization — normalize study/series descriptions across sites and scanners for consistent downstream processing
- HL7 FHIR notifications — notify hospital EMR/RIS systems when studies are de-identified, approved, or exported
- Reversible de-identification — organization-owned keys for internal use cases where re-identification may be required
- PACS/VNA query-retrieve — pull studies from hospital archives on demand rather than waiting for push
- Multi-tenant SaaS deployment — shared infrastructure with per-organization data isolation for radiology groups managing multiple facilities

Cost Estimate (Development / Proof of Concept)

Resource	GCP Monthly	AWS Monthly
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Containers (Cloud Run / ECS Fargate)	~\$5–15	~\$5–15
PostgreSQL (Cloud SQL / RDS)	~\$10	~\$15
Object Storage (GCS / S3)	~\$0.02/GB	~\$0.02/GB
Load Balancer	Included	~\$16 (ALB)
Total (development)	~\$20–40/month	~\$40–60/month

Production costs scale with data volume. A 1,000-session multi-site study (~500 GB) would cost approximately \$50–100/month in storage on either cloud.

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AEGIS

Secure, browser-based medical image de-identification and sharing for any DICOM modality — serving both multi-site research teams and hospital radiology departments from a single multi-cloud platform, with automated defacing, pixel-level PHI detection, and administrator review before any data is released. Runs on GCP, AWS, or Azure.